

version: 1.0

VSTProfiler is an open-source software project dedicated to the digital cloning of hardware amplifiers. Developed using the **JUCE framework**, that can operates as both a standalone application and a plugin (VST3/AU). The project focuses on high-fidelity sonic reproduction, **ultra-low latency** performance, and a robust architecture designed for community contribution. It integrates a professional workflow including multi-platform CI/CD (Windows, macOS, Linux), real-time performance metrics, and comprehensive technical documentation.

The following roles are involved in the beta testing phase.

Role Name	Description
End User	Musicians or producers using the plugin for real-time audio processing without perceptible delay.
Contributor	External developers seeking to understand the architecture, submit bug fixes, or improve the DSP engine.
QA Engineer	Internal team members responsible for validating metrics (CPU/Latency), unit tests, and CI/CD pipeline health.

All features listed below will be demonstrated during the beta defense.

Feature ID	User role	Feature name	Short description
F1	End User	Create Digital Amp Copy	Analyzes your real amplifier and creates a digital version that sounds just like it
F2	End User	Instant Response (No Delay)	Ensures you hear sound immediately when you play no lag or delay
F3	End User	Save Amplifier Sound	Saves your amplifier's sound as a file you can use again or share with others
F4	End User	Load Amplifier Sound	Opens a saved amplifier sound file to use that specific tone
F5	End User	Save Complete Session	Saves your entire current setup (which amp, all settings, knob positions) as a preset
F6	End User	Load Complete Session	Opens a saved preset to restore all your previous settings exactly as they were
F7	End User	Save Personal Preferences	Remembers your favorite settings (colors, performance options, etc.) between sessions
F8	End User	Load Personal Preferences	Automatically applies your saved preferences when you open the plugin
F9	End User	Control Interface	Visual controls (knobs and switches) to adjust your amplifier settings
F10	End User	Audio Level Display	Shows real-time visual meters so you can see your audio levels
F11	End User / Contributor	Simple Installation	Easy install process for using the plugin or contributing to development
F12	Contributor	Technical Guide	Detailed documentation explaining how the technology works

Feature ID	Contributor	Feature name	Comments	Short description
F14	Contributor	Open Source	Ease of Contribution	Public access to all code with guidelines for contributing improvements (README, license, contribution guidelines)
F15	QA Engineer	Automated Build System		Automatic testing that checks if the software works on Windows, Mac, and Linux
F16	QA Engineer	Tests and Coverage		Automatic unit tests for funtion of the project and automatic coverage
F17	QA Engineer	Performance Data		Files that track how well the software performs for analysis
F18	QA Engineer	Error Tracking System		Captures technical details and errors to help fix problems
F19	End User / QA Engineer	Optional Error Reporting		Lets users choose to share their hardware details in logs to help improve the software
F20	QA Engineer	Error Alert Messages		Pop-up warnings that appear when something goes wrong

| F21 | QA Engineer | Automated Quality Checks | Automatic tests that verify the audio processing works correctly |

4. Success criteria

Feature ID	Key success criteria	Indicator/metric	Result
F1	Sound Accuracy	Digital copy sounds almost identical to the real amplifier	Pending
F2	Instant Response	Sound delay is less than ~10 milliseconds (imperceptible to humans)	Achieved
F3	Save Amplifier Sound	File is created successfully with all sound characteristics included	Pending
F4	Load Amplifier Sound	Loaded sound matches the saved version with 100% accuracy	Pending
F5	Save Complete Session	File is created successfully with all the states for elements of the plugin included	Pending
F6	Load Complete Session	Opens a saved state of the plugin to restore it to how it was before last closing it	Pending
F7	Save Personal Preferences	File is created successfully with all the settings (colors, performance options, etc.) included	Pending
F8	Load Personal Preferences	Automatically applies your saved settings when you open the plugin	Pending
F9	Control Interface Smoothness	Interface runs smoothly at 60 frames per second without affecting audio	Achieved
F10	Audio Level Display	Visible UI element showing real-time visual meters of the audio levels	Pending
F11	Simple Installation	A new developer can install and build the project in under 15 minutes	Achieved
F12/F13	Code Quality Standards	All code meets formatting standards and passes quality checks	Pending
F14	Open Source	The project repository is open source	Achieved
F14	Ease of Contribution	The project repository and/or its wiki contains a README LICENSE and CONTRIBUTING files	Achieved
F15	Automated Build System	Software builds successfully on Windows, Mac, and Linux at the same time	Achieved
F16	Tests and Coverage	The projects CI/CD performs unit tests for funtion of the project and automatic coverage	Pending
F17	Performance Data	Performance data file are generated at runtime if the setting is enabled	Pending
F18	Error Tracking System	Errors warnings and other debug messages are generated and stored in a log file	Achieved

Feature ID	Optional Error Reporting Key Success Criteria	Existence of an opt in setting in the plugin to enable identifiable info in logs Indicator/metric	Pending Result
F20	Error Alert Messages	Toggleable and configurable popups for logs to apears in plugin's UI	Pending
F21	Automated Quality Checks	Presence of CI/CD element to check sound accuracy	Pending